Cardiovascular Diseases and Metabolic Syndrome

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Elisa Roldan Galvan, John A. Snowden, and Diana M. Greenfield

55.1 Introduction

Cardiovascular disease (CVD) remains the leading cause of death globally, with 17.9 million deaths each year, mainly for coronary heart disease and cerebrovascular disease. The World Health Organization (WHO) estimates that the largest increment in deaths has been for ischemic heart disease, rising by more than 2 million in 2000 to 8.9 million in 2019, representing 16% of all global deaths. It is predicted that by 2030, almost 23.6 million people will die from CVD (WHO 2023).

Moreover, there is an increased incidence of metabolic syndrome (MetS), with reported prevalence rates of 31–49% after transplant.

After HCT, either autologous or allogeneic, an increased incidence of CVD has been shown compared with the normal population and sibling donors (Baker et al. 2007). Retrospective EBMT analyses have shown the cumulative incidence of a first cardiovascular event 15 years after HCT rises to 6%.

55.2 Cardiovascular Disease in HCT

55.2.1 Abdominal Obesity

Abdominal obesity measured by waist circumference represents fat accumulation (visceral adipose deposits) which independently confers cardiometabolic risk. Changes in waist circumferences are seen after HCT with, for example, corticosteroid use and with onset of sarcopenic obesity.

55.2.2 Dyslipidemia

Dyslipidemia is defined by elevated levels of total cholesterol, LDL-C, or triglycerides or low levels of HDL-C. Prevalence in general population is estimated at 25% in the USA (Baker et al. 2007) and in European countries. Evidence suggests allo-HCT recipients have significantly higher risk of new onset dyslipidemia (RR 2.1 95% CI 1.15–4.65) compared with auto-HCT (Tichelli et al. 2007a) with the prevalence post-HCT estimated to be 43–73% (Chow et al. 2014; Oudin et al. 2015). TBI significantly increases the risk of dyslipidemia.

E. R. Galvan (✉) · J. A. Snowden
Department of Haematology, Sheffield Teaching Hospitals NHS Foundation Trust, Sheffield, UK

Division of Clinical Medicine, School of Population Health, School of Medicine and Population Health, University of Sheffield, Sheffield, UK
e-mail: Elisa.roldangalvan1@nhs.net; john.snowden1@nhs.net

D. M. Greenfield
Division of Clinical Medicine, School of Population Health, School of Medicine and Population Health, University of Sheffield, Sheffield, UK

Specialised Cancer Services, Sheffield Teaching Hospitals NHS Foundation Trust, Sheffield, UK

Department of Haematology, Sheffield Teaching Hospitals NHS Foundation Trust, Sheffield, UK
e-mail: diana.greenfield@nhs.net

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55.2.3 Hypertension (HTN)

Hypertension (HTN) in the general population is defined as systolic BP ≥140 mmHg or diastolic BP ≥90 mmHg but defined in context of MetS as systolic BP ≥135 mmHg or diastolic BP ≥85. HTN in people following allo-HCT is 2.06 times (95% CI 1.39–3.04) more likely compared with sibling donors or auto-HCT (Baker et al. 2007).

55.2.4 Insulin Resistance or Diabetes Mellitus (IR/DM)

DM is characterized by hyperglycemia resulting from defects in insulin secretion (type 1) and insulin resistance (type 2), and it is defined as a fasting pGL ≥7 mmol/L, an HbA1C ≥6.5%, a 2-h plasma glucose ≥11.1 mmol/L during a glucose tolerance test (GTT), or a random glucose ≥11.1 mmol/L.

Both allo-HCT and auto-HCT recipients have been found to report DM more often than sibling donors (OR for allo-HCT, 3.65; 95% CI, 1.82–7.32; OR for auto-HCT, 2.03; 95% CI, 0.98–4.21) (Baker et al. 2007). High-dose corticosteroids (cumulative PRD dose of >0.25 mg/kg/day) increase the likelihood of developing DM (RR, 3.6; 95% CI, 1.7–7.5) and for having persistent DM at 2 years post-HCT (RR, 4.1; 95% CI, 1.0–18.2) (Majhail et al. 2009a, b). TBI is also a well-evidenced risk factor (Hirabayashi et al. 2014). TBI was linked to 3.42 times greater risk of diabetes development.

55.3 Risk Factors for CVD in HCT Recipients

There are several well-defined risk factors which indicate increased susceptibility to post-transplant CVD (Table 55.1).

55.3.1 Contribution of Type of Transplant

The contribution of the type of transplant in the CVD developing is not well defined. In a previous EBMT study, the cumulative incidence for the first CV event was 7.5% at 15 years post-allo-HCT versus 2.3% post-auto-HCT (Tichelli et al. 2007a). However, a more recent cross-sectional multicenter study found no difference in auto- vs allo-HCT (Greenfield et al. 2021).

A higher incidence of early cardiac events (ECE), occurring within the first 100 days after HCT, has been reported in PTCy-based approaches. Older age, sequential conditioning regimen, and CY exposure before transplant are additional risk factors for ECE in this context (Duléry et al. 2021).

Table 55.1 Risk factors for cardiovascular disease

Patient's baseline cardiovascular (CV) risk factors	Family history Female sex Smoking Hypertension Dyslipidemia Diabetes Obesity Sedentary habit Chronic kidney disease
Chemotherapy strategies used in the underlying disease	Anthracyclines (doxorubicin or equivalent >250 mg/m^2) Steroids Site-specific radiotherapy—Dose >35 Gy to a field including the heart or >15–35 Gy + doxorubicin >100 mg/m^2 Total body irradiation Other cardiotoxic agents: cyclophosphamide, busulfan, melphalan, thiotepa
GVHD prophylaxis	Steroids Calcineurin inhibitors
GVHD	Grades II–IV aGvHD Any grade cGvHD
Other secondary late effects	Hypogonadism Premature menopause Hypothyroidism

Chow et al. (2014); Oudin et al. (2015); Lopez-Fernandez et al. (2021); Greenfield et al. (2021)

55.4 Metabolic Syndrome Definition

Metabolic syndrome (MetS) is a cluster of interrelated factors which increase the risk of cardiovascular disease, diabetes mellitus (DM), and all-cause mortality. MetS is defined as the presence of three out of five risk factors as follows:

- Abdominal obesity measured by waist circumference: with population- and country-specific definitions
- Triglycerides ≥1.7 mmol/L
- HDL-C (men) <1.0 mmol/L
- HDL-C (women) <1.3 mmol/L
- Blood pressure ≥130/≥85 mmHg
- Fasting glucose ≥5.6 mmol/L

The International Diabetes Federation (IDF) estimates 25% of the world's population has MetS. After HCT, there is an increased incidence of MetS, with reported prevalence rates of 31–49% (Majhail et al. 2009b; Oudin et al. 2015; Greenfield et al. 2018). More recently, Greenfield et al. did a cross-sectional, multicenter, noninterventional study of 453 adult HCT patients surviving a minimum of 2 years post-transplant (Greenfield et al. 2021). Overall prevalence of MetS was 37.5%, rising to 53% in patients above 50 years old. Of significance, higher occurrence of cardiovascular events was found in those patients with MetS than in those without MetS (26.7% versus 9%, $p < 0.001$).

55.4.1 The Immune System as a Mediator of Metabolic Syndrome

Extensive research is now recognizing obesity as a chronic inflammatory state which drives the development of insulin resistance and MetS. Body mass index and the degree of abdominal adiposity are positively correlated with serum inflammatory markers, including C-reactive protein, TNF-alfa, and IL-6. In addition, adipocytes are now known to serve critical endocrine and immune function. Within lean individuals, adipose tissue is predominantly populated by regulatory T cells (Tregs), Th-2 cells, and M2 macrophages, whereas, in obese individuals, adipose tissue contains Th 1, M1 macrophages, CD8+ T cells, B cells, and dendritic cells. In the setting of HCT, significant tissue injury, inflammation, and perturbations in immune cell number and function occur, and this appears to play a key role in the initiation and development of the process (Turcotte et al. 2016).

55.5 Preventative Practices in the HCT and Late-Effect Clinic: A Practical Approach

The fact that HCT survivors require close follow-up and clinical review provides an opportunity to deliver screening for late effects and other long-term consequences of treatments. Screening for cardiovascular risk factors, including MetS and CV events, can be integrated into a broader program of long-term and late-effect follow-up. If cardiovascular risk factors are detected, they can usually be referred back to primary care clinicians or facilitate the referral to the appropriate specialist department. Algorithms, such as the Framingham risk score, may be useful in estimating a person's projected risk of developing CVD in the general population. Although these CVD risk scores have not been validated in HCT survivors and may potentially underestimate the risk, they may be reasonable to use as an initial guideline. However, there should be direct referral for clinically urgent cardiovascular problems to relevant hospital specialists, and, ultimately, a close communication between the different clinicians involved in the care of the patient is indispensable.

Given the specialized complexity of HCT and its many complications, the HCT clinic and associated late-effect service can have a major role in coordinating care and facilitating communication between other relevant specialists. This aspect is underpinned by the FACT-JACIE standards which feature systematic provision for late-effect follow-up, including cardiovascular risk factors and complications (FACT-JACIE eighth edition).

For the HCT late-effect clinic, Table 55.2 has been published as a guide to facilitate screening in the EBMT-CIBMTR guidelines (DeFilipp et al. 2017). This is a consensus opinion, and there is no good evidence of the safety or clinical effectiveness of these recommendations in HCT patients. Based on the available evidence, it is important to screen for other factors in HCT patients, including (a) personal history, (b) family history, (c) type of transplant (allo or auto), (d) use of TBI, (e) history of acute or chronic GvHD, and (f) use of CNI (CSA, TAC) (DeFilipp et al. 2017).

Table 55.2 Screening guidelines for metabolic syndrome and cardiovascular risk factors for adult and pediatric patients among the general population and HCT survivors (taken from DeFilipp et al. 2017)

	General adult population	Adult long-term HCT survivors (Majhail et al. 2012)	General pediatric population	Pediatric long-term HCT survivors (Pulsipher et al. 2012)
Weight, height, and BMI	Weight, height, and BMI assessment in all adults (no specific recommendation for screening interval)	No specific recommendations	Weight, height, and BMI assessment after 2 years of age (no specified screening interval)	Weight, height, and BMI assessment yearly
Dyslipidemia	For persons with increased risk for coronary heart disease, assessments should begin at age 20	Lipid profile assessment every 5 years in males aged ≥35 years and females aged ≥45 years	Lipid panel between 9 and 11 years of age or earlier if family history	Lipid profile at least every 5 years; if abnormal, screen annually
	The interval for screening should be shorter for people who have lipid levels close to those warranting therapy and longer intervals for those not at increased risk who have had repeatedly normal lipid levels	Screening should start at age 20 for anyone at increased risk (smokers, DM, HTN, BMI ≥30 kg/m^2 and family history of heart disease before age 50 for male relatives or before age 60 for female relatives)		
Blood pressure	Blood pressure assessment every 3–5 years in adults aged 18–39 years with normal blood pressure (<130/85 mmHg) who do not have other risk factors	Blood pressure assessment at least every 2 years	Blood pressure assessment yearly after the age of 3 years, interpreted for age/sex/height	Blood pressure assessment at each visit and at least annually
	Blood pressure assessment annually in adults aged ≥40 years and for those who are at increased risk for high blood pressure (blood pressure 130–139/85–89 mmHg, those who are overweight or obese, and African–Americans)			
Hyperglycemia	Screening for abnormal blood glucose (HbA1C, fasting plasma glucose, or oral glucose tolerance test) every 3 years in adults aged 40–70 years who are overweight or obese	Screening for type 2 DM every 3 years in adults aged ≥45 years or in those with sustained higher blood pressure (>135/80 mmHg)	Fasting glucose every 2 years after the age of 10 years in overweight children with other risk factors	Fasting glucose at least every 5 years; if abnormal, screen annually

BMI body mass index, *DM* diabetes mellitus, *HbA1C* hemoglobin A1C, *HCT* hematopoietic cell transplantation, *HTN* hypertension

55.6 Future Directions: Implementation, Education, and Research

As survival after HCT gradually increases, there is also increased recognition of HCT-related CVD and its risk factors, including MetS. Most research has been cross-sectional and observational. There are progressively more information in the pathogenesis of Mets, but still more prospective research is needed both on defining the incidence above the normal aging population and on interventional strategies, targeting individual risk factors and/or components of the MetS.

New indications for HCT, such as systemic autoimmune disease, and alternative platforms, such as haplo-HCT or cord-HCT, require individualized assessment. Pharmacological, lifestyle, and rehabilitation interventions are common in the general population in respect to CVD. However, their impact in HCT recipients (both before and after HCT) needs to be defined in the context of the wide range of indications and age at which patients receive their HCT, along with the individual prognosis of each indication after successful HCT.

Key Points

- Long-term survivors have an increased risk of premature MetS and CVD. The best approach is to screen all patients (i.e., both autologous and allogeneic HCT) according to international consensus guidelines (DeFilipp et al. 2017) and manage risk factors on an individual basis.
- The challenge of universal implementation of screening and management of late effects across various health services providing HCT will be facilitated by FACT-JACIE accreditation standards.
- Randomized controlled trials of interventional strategies and mechanistic studies of cardiovascular risk in HCT survivors are still needed.

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